

##

Data Scheme for StarGate

- This document **only defines** the farming performance gauge data of the StarGate bridge.
- This document does not specify optimized algorithms that collect and compute the data.

1. Snapshots at a given time moment

- **STGpB~c~**: STG-per-Block on the chain c.
- **ALLOC~c,p~**: AllocPoint for the pool p on the chain c.
- **T_ALLOC~c~**: Total AllocPoint on the chain c.
- **USDpLP~c,p~**: USD-per-LP for the pool p on the chain c.
- **STK~c,p~**: Staked LP Tokens for the pool p on the chain c.
- **STGP**: STG price quote in USD
- **STG_ACC~c,p~**: ACC reward that would have been accumulated for one unit of LP since the beginning of farming if the unit had been kept staked throughout, on the pool p on the chain c.

2. Block-level data

- **B-Average** of feature X in a block is defined to be $\frac{1}{n} * \sum_{i=0}^{n-1} x_i$, where $\{x_i \mid i = 0, 1, \dots, n-1\}$ are values that are held by X sequentially in the block and $x_k \neq x_{k+1}$ for all k .
Example: If Balance has a constant value 30 in the block, then the B-Average of Balance is 30. If Balance has two values 30 and 40 in total sequentially in the block, then the B-Average of Balance is $\frac{1}{2} * (30+40) = 35$.
- **STGpB^b~c~, ALLOC^b~c,p~, T_ALLOC^b~c~, USDpLP^b~c,p~, STK^b~c,p~, and STGP^b~**: The B-Average of relevant snapshot feature on a given block.
- **STGpLP^b~c,p~**: STG-rewards-per-LP on a given block, equal to the increment of STG_ACC~c,p~ from that block's opening time to the next block's.
- **ROI^b~c,p~**: Return-On-Investment on a given block, equal to the increment of USDpLP~c,p~ * STG_ACC~c,p~ from that block's opening time to the next block's.
- **APY^b~c,p~**: Annual Profit Yield on a give block, equal to $(1 + ROI^b~c,p~)^{**} (\text{year} / \text{AvgBlockTime}~c~) - 1$, where AvgBlockTime is the average time elapse that a block is minted.

3. Timeframe-level data

- **Timeframe:** time interval $[T_a, T_b]$, where T_a and T_b are the Unix timestamp of an integer second. Note T_b does *not* belong to the interval.
- **A block belongs to a timeframe** if and only if the timestamp of the block belongs to the timeframe.
- **BlockValue(X, B)** denotes the value of a feature X on the block B. **An examples** of BlockValue(X, B) is the B-Average.
- Some derived features around a feature X on a timeframe are defined as follow:
 $\#-Sum = \sum_{i=0}^{n-1} BlockValue(X, B_i)$,
 $\#-Product = \prod_{i=0}^{n-1} BlockValue(X, B_i)$,
 $\#-Average = 1/n * \sum_{i=0}^{n-1} BlockValue(X, B_i)$,
 where $\{B_i \mid i = 0, 1, \dots, n-1\}$ are all blocks that belong to the timeframe.
- $STGP^{a \sim c \sim}, ALLOC^{a \sim c, p \sim}, T_ALLOC^{a \sim c \sim}, T_ALLOC^{a \sim c \sim}, SLP^{a \sim c, p \sim},$
 $USDpLP^{a \sim c, p \sim}, STGpLP^{a \sim c, p \sim}, STK^{a \sim c, p \sim},$ and $STGP^{a \sim c}$: The #-Average of relevant block-level feature on a timeframe.
- $STGpLP^{s \sim c, p \sim}$: The Sum of $STGpLP^{b \sim c, p \sim}$, on a timeframe, for the pool p on the chain c.
- $ROI^{s \sim c, p \sim}$: The Sum of $ROI^{b \sim c, p \sim}$ on the timeframe for the pool p on the chain c.
- $APY^{s \sim c, p \sim} := (1 + ROI^{s \sim c, p \sim})^{** (year / FrameTime)} - 1$, where FrameTime is the length of timeframe.